

INTERN FINAL PRESENTATION

A DevOps Summer Recap



Trustan Price · DevOps / CyberSecurity Intern · ALI/Services · August 2026

FROM NOTEBOOK TO PIPELINE

Twelve weeks of a DevOps intern summer from start to finish.

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Where I Started

Same instinct for operational visibility, aimed at a very different problem this time.



Primary Goal

- Last summer: migrate AWS CloudWatch dashboards to Grafana
- This summer: build a CI/CD workflow to deploy AWS resources using custom actions, base templates, and reusable workflows



Core Structure

- YAML both summers
- Last summer: Azure DevOps (AzDO)
- This summer: GitHub Actions
- Also learning AWS Cloud from the ground up



Security & Reliability

- OIDC and role ARNs in place of hardcoded keys
- Researched DevOps scanning tools to better understand their capabilities.
- Received mentorship on SAST scanning, Dependabot, and dependency management best practices.

Workflow anatomy: workflows → jobs → steps → runners, base templates

Building the Foundation

Weeks 4-6



CloudFormation Templates

Parameters, Resources, Mappings, and Outputs — modeling dependencies and managing drift across environments.



OIDC Trust Relationships

GitHub as identity provider, a role trust policy scoped to org/repo, and short-lived token exchange instead of long-lived secrets.



Security & Dependency Scanning

Gained exposure to SAST scanning using DevOps Tool Scan (CodeQL), along with Dependabot and dependency management best practices.



GitHub Actions Architecture

Event → Workflow → Job → Runner → Steps → Actions
Reusable Workflows + Custom Actions

AWS Cloud Practitioner, In Practice

Same four domains — studied for the exam, then put to work on the platform itself.



Cloud Concepts

Chose between EC2, Elastic Beanstalk, App Runner, and Lambda based on how much infrastructure control a given workload actually needed.



Security and Compliance

Put the shared responsibility model into practice — OIDC trust relationships and least-privilege IAM on my side, AWS's managed patching on theirs.



Cloud Technology and Services

Worked hands-on with S3, Lambda Functions, and Secrets Manager



Billing, Pricing, and Support

Tracked spend with Cost Explorer and Budgets, and compared Savings Plans against Reserved Instances for the platform's support tier.

Mid-Summer & the Hackathon

Weeks 7 and 8 — the mid-summer presentation and the Caterpillar Global Hackathon.



Delivering the Mid-Summer Story

Manager feedback shaped how — and whether — the technical work got told.

- CloudFormation templates tested and validated end-to-end ahead of the demo
- 1:1 with Vinit on how to structure the story and what to highlight
- Delivered “Beyond the Script” — the case for agentic DevOps at Caterpillar
- AWS practice exam scores climbing into cert-ready range
- Contributed to the development of both a web and mobile application for a hackathon
- Shifted my mindset from solely learning to balancing both learning and development

THE FEEDBACK

"Grow vertically in your area of ownership while expanding horizontally across the DevOps team to build a well-rounded understanding of the organization."

The presentation itself became a deliverable — not just the pipeline behind it.



Team 9 — 20/20 VisionLink

A recommendation-driven dashboard prototype, built in a single week.



My Role

- Product manager — ideated the project from scratch
- Presented the business case to get the team on board
- Built the bulk of the app's functionality — the recommendation system itself



The Team & Stack

- Eric: frontend · Anokhi: ML backend
- Web: React + TypeScript
- Mobile: native, built in Xcode
- Backend: Ollama
- UI in Figma Make + AI orchestration



The Build

- Recommendation-based dashboard tracking user interactions
- Secured by two-factor authentication (2FA)
- Alerts fleet managers, Google-style, if someone else views their assets

CI/CD Platform & the Pivot

Weeks 9 and 10 — the first working demo, and the pivot into research of my own.



First Demo → First Real IAM

Deploying three resources end-to-end — and finally understanding the permissions underneath.



The First Demo

- Finished and presented the CI/CD pipeline's first real demo
- An end-to-end deployment of three AWS resources
- Utilized base templates, custom actions, and reusable workflows



Reverse-Engineering IAM

- Learned to read and reverse-engineer IAM roles directly
- Understanding policies and permissions instead of just following a template
- Finding the difference between CFN and OIDC roles

The moment the pipeline stopped being copied from a template and started being understood.

From Mentorship to My Own Research

Where the seed of a summer's mentorship started becoming something of my own.

- Weeks 1–9:
 - Built understanding of GitHub Actions, CloudFormation, OIDC, IAM
 - Carrying lessons forward from last summer's AIOps work
- Week 10:
 - Began implementing Genesis - the parallel with my original LLMOps pipeline research
 - Layered directly onto that same CI/CD workflow
- Not a new project: my initiative to enhance the existing workflow
 - I wanted to find a way to positively influence the team through a future-focused initiative that could provide long-term value beyond my internship.
 - This research ultimately became the foundation for my Emerging Technology Initiative.

The seed of a summer of mentorship, starting to bloom into research of my own.

What's Next

The people behind the summer, the ask for what comes after, and a look back at the whole journey.



Emerging Tech: My Influence

Sherlock, InfraGen, Wiz, and Genesis — where each one leads, in my own words.

Sherlock

- AI-powered troubleshooting assistant that ingests pipeline or CloudFormation failure context, analyzes logs and events, and identifies likely root causes.
- Best positioned as advanced AIOps: observe, diagnose, and recommend after failures occur.

InfraGen

- Multi-agent infrastructure engineering platform that uses Bedrock agents, knowledge bases, and GitHub automation to generate CloudFormation artifacts from design inputs.
- Best positioned as Agentic DevOps / emerging LLMOps: plan, generate, propose changes, and hand off at the approval and deployment gate.

Wiz

- Cloud security platform that identifies live cloud risks, maps affected resources, providing AI-assisted remediation guidance.
- Supports shift-left Infrastructure-as-Code scanning across CloudFormation, Terraform, ARM, and Bicep with severity-based findings, secrets detection, and policy reporting.
- Provides governance capabilities including issue lifecycle tracking, time-bound exceptions, dashboard integration, SLA monitoring, and CI/CD security reporting.

Genesis

- LLMOps workflow that evaluates deployment changes, assesses risk, and routes actions through PROCEED, AUTO-FIX, or REVIEW-PR paths.
- Combines proactive scanning, governed remediation, observability, and AI-driven recommendations to improve deployment safety.
- Enhances Security, CloudOps, and FinOps through faster feedback, automated reviews, and human-governed decision support.

Built Together Not Independently

The mentorship that shaped the technical decisions all summer.



Deeksha - Mentor

- Explained LTA, Dependabot, Dependency Management and SAST scanning
- Covered custom actions, reusable workflows, CI/CD, and base templates



Vinit - Teammate

- Deployment and repo structure
- The deep dives that shaped CloudFormation and release design



Nathan - Manager

- Gave direction on learning vertically
- At each stop, branched out horizontally into CloudOps, FinOps, and Security



Cora - Hack Mentor

- Hackathon mentor — guided Team 9 to a second-round appearance
- Walked me through her day-to-day in FinOps

Also Sonia, Angie, and Rob — OIDC + HR Advice

Where I Want to Take This

“My goal in this internship is to understand where the team is today, identify where I can add real value, and execute on that with forward-looking approaches the team has not yet used, so I can contribute not just as an intern, but as someone helping push the work toward what comes next.”

Vision

A path that builds on this summer's CI/CD work and the Genesis research, not a restart.

Impact

Fewer manual pipeline hours, safer deployments, a stronger security posture.

Continuity

Day-to-day platform work feeding directly into the bigger picture.



From Notebook to Deployment



Week 1

Onboarding



Week 4

Foundation



Week 7

Mid-Summer



Week 8

Hackathon



Week 10

Genesis Pivot

Twelve weeks of GitHub Actions, CloudFormation, and disaster-recovery thinking — turned into a working, documented CI/CD platform.

Trustan · Questions welcome